



SURF Student Colloquium

NIST – Gaithersburg, MD

August 13-15, 2024

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Major: Computer Science w/ Statistics minor

**Academic Standing
(Sept. 2024):** Sophomore

**Future Plans
(School/Career):** Software Engineering / Cybersecurity

**NIST Laboratory,
Division, and Group:** Engineering Laboratory, Materials and Structural Systems Division, Community Resilience Group

**NIST Research
Advisor:** Dr. Dylan Sanderson

Title of Talk: Verification and Validation of Computational Models for Analyzing Future Impacts of Sea Level Rise

Abstract:

Community resilience is defined as the ability of communities to prepare for anticipated hazards, adapt to changing conditions, and withstand and recover rapidly from disruptions. Sea level rise poses significant threats to coastal communities as it has the potential to negatively impact both infrastructure and households. Given this threat, new computational models for community resilience planning are needed to help communities prepare for future hazards. To better understand and predict how sea level rise will impact future communities, I have been working on the verification and validation (V&V) of an agent-based model (ABM) of household adaptation to sea level rise. In this ABM, agents respond to future exposure to sea-level rise and make decisions regarding adaptation using reinforcement learning. Galveston, Texas is used as a testbed community as it is potentially subject to over 2 feet of sea level rise in the next 100 years. Multiple V&V tests were performed such as comparison to other models and sensitivity analysis. Regarding comparison to other models, I evaluated three different agent decision-making mechanisms: (1) agents consider their exposure to sea level rise, (2) agents use a reward signal, and (3) agents use utility theory and Cobb-Douglas utility functions. Sensitivity tests using the reinforcement learning model were then performed and showed how the agent decision-making process is sensitive to the agent's reward signal. The results demonstrate the difference between two different ABMs, one that uses reinforcement learning and one that does not, along with representing how both predict agent behavior.